

Prompt Confusion Phase 04 Smoke Findings

Prompt-EOS smoke memo for the grouped conflict probe run on April 10, 2026.

MAIN READ

The prompt_eos smoke path is operational end to end, and the grouped Phase 04 conflict signal is still clearly decodable after fixing the prompt-end pooling target.

Run Summary

Capture run

86287b7c78a3

48 of 48 smoke examples captured.

Analysis run

f844453d2c91

Grouped workflow_label probe completed.

Evaluation

**24
groups**

matched_pair_id
grouped 5-fold evaluation.

Pooling

prompt_eos

Prompt-side EOS token instead of raw last token.

This smoke run uses the updated Phase 04 workflow snapshot with:

- capture.pooling = "prompt_eos"
- analysis.pooling = "prompt_eos"
- analysis.group_column = "matched_pair_id"
- capture.add_generation_prompt = false

Accuracy Table

Layer	Accuracy Mean	Balanced Accuracy	Std Dev	Selectivity
0	0.770	0.770	0.0748	+0.375
12	0.685	0.685	0.0943	+0.105
24	0.785	0.785	0.0889	+0.265

36	0.820	0.820	0.1166	+0.325
44	0.725	0.725	0.0922	+0.250

Smoke interpretation

The best grouped readout in this smoke run is layer 36 at 0.820 balanced accuracy. Layer 24 and layer 0 are also strong, while layer 12 weakens materially. On a 48-example smoke slice, this is enough to say the Phase 04 signal survives the prompt-end pooling correction, but not enough to headline a stable geometry claim yet.

What Changed

Two capture-path fixes matter for this run:

- pooling now anchors on the last prompt-side EOS token instead of the final rendered newline
- vLLM router capture now accumulates logits across prefill chunks rather than overwriting the buffer with the final chunk only

Those changes remove the earlier router indexing failures and make the `prompt_eos` feature meaningfully aligned with the intended prompt boundary.

Stability Note

The major operational bug from the previous smoke run is fixed: the capture job now completes all 48 rows instead of dying mid-run with router indexing errors.

There is still one stability concern worth treating as active:

- the capture worker has been finishing successfully, but Modal/vLLM still emits `EngineCore_DP0 died unexpectedly during teardown`

The most obvious risk factor in the old configuration was that capture still forced `enable_chunked_prefill = false` even though Qwen3 MoE warns against that mode. That has now been patched so capture workers use chunked prefill by default in the workflow orchestrator and related DX terminal copies. A fresh smoke rerun on capture run `acf299a8c4ae` confirms that the old unsupported-mode warning is gone.

I also added explicit worker teardown that calls `llm_engine.shutdown()` and clears CUDA caches. A second rerun on capture run `c8911ffa6b50` still reports `EngineCore_DP0 died unexpectedly during shutdown`. So the remaining issue is not prompt capture correctness and not the old unsupported prefill mode; it is a post-success worker teardown problem.

Next Step

The immediate next step is:

1. isolate why vLLM workers still die noisily on shutdown after a successful flush, most likely by reducing worker fan-out, shrinking worker reuse, or changing how Modal containers retire after the local entripoint completes
2. once teardown is clean enough to trust operationally, run the full 288-row Phase 04 recapture with `prompt_eos`
3. only after the full recapture finishes, read probe quality under grouped evaluation before making any stronger representational claim

The main methodological point is unchanged: this smoke result earns a full rerun, not a mechanism claim.

Prompt Confusion Phase 04 Smoke Findings - accuracy summary from capture run 86287b7c78a3 and analysis run f844453d2c91, with stability follow-up from capture runs acf299a8c4ae and c8911ffa6b50.